CSCI 6850

Graph Machine Learning Paper Reading and Presentation Assignment

Objective

The goal of this assignment is to explore recent advances in **Graph Machine Learning** by identifying and presenting a research paper in the field. Students will work in small groups to analyze and critically evaluate the paper, synthesizing key concepts and findings to share with the class.

Assignment Overview

- **Group Size**: Up to 3 students per group
- **Topic**: A research paper related to **Graph Machine Learning**
- **Presentation Duration**: **30 minutes per group**
- **Presentation Date**: TBD. Will be in weeks 13-15.

Instructions

1. **Paper Selection**
   - Choose a research paper in the domain of **Graph Machine Learning**.
   - Papers must be top AI conferences (e.g., NeurIPS, ICML, ICLR, AAAI) or reputable journals (Impact factor >=8.0).

2. **Presentation Guidelines**

   Your 30-minute presentation should include the following:
   - **Motivation & Problem Statement**
     - What problem does the paper address?
     - Why is it important in the context of deep learning and GML?
   - **Related Work**
     - Briefly discuss previous approaches to the problem.
   - **Methodology**
     - Explain the key ideas and techniques introduced in the paper.
     - Discuss the **graph neural network (GNN) architecture** or other relevant methods used.
   - **Experiments & Results**
     - How was the model evaluated?
     - What datasets were used?
     - Key performance metrics and comparisons to baselines.
   - **Critical Analysis**
     - Strengths and weaknesses of the paper.
     - Potential applications or limitations.
   - **Future Work & Discussion**
     - What are the next steps or open questions in this research area?
   - **Q&A Session** (5 minutes)

3. **Presentation Format**
   - Each group must prepare **slides**.

- Use **diagrams, visualizations, and examples** to make your presentation engaging.
- Keep text minimal—focus on explaining concepts clearly.

4. **Submission Requirements**
    - Submit your **slides (PDF format)** at least **one day before** your presentation.
    - Include a **one-page summary** covering key points from the presentation.

Resources

- Recommended venues for finding research papers:
    - arXiv (Preprints)
    - Google Scholar
    - OpenReview (ICLR)
    - Papers With Code

Additional Notes

- **Avoid selecting outdated or overly simple papers**—choose something recent (last 5 years) and impactful.
- **Collaborate effectively within your group**, ensuring equal contributions.
- **Practice your presentation** to stay within the time limit.

Grading Criteria (Total: 100 points)

Criteria	Points	Description
Paper Selection	10	Quality and relevance of the selected paper.
Clarity & Organization	20	Logical flow of presentation, clarity of explanations.
Depth of Understanding	25	Demonstrated comprehension of methods and results.
Critical Analysis	20	Strengths, weaknesses, and discussion of future work.
Engagement & Delivery	15	Quality of presentation, visuals, and group participation.
Q&A Handling	10	Ability to answer questions effectively.